```
# Install (run once)
!pip install torch transformers datasets peft accelerate

# Import libraries
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

```
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
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Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hugg
```

```
model_name = "distilbert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
config.json: 100%  483/483 [00:00<00:00, 12.0kB/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

```
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 739B/s]

vocab.txt: 232k/? [00:00<00:00, 3.11MB/s]

tokenizer.json: 466k/? [00:00<00:00, 6.18MB/s]

model.safetensors: 100%  268M/268M [00:06<00:00, 69.8MB/s]

Loading weights: 100%  100/100 [00:00<00:00, 119.94it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
```

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
classifier.bias          | MISSING    |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |
classifier.weight        | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
dataset = load_dataset("csv", data_files="/content/sentiment_dataset.csv")

def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")

tokenized_dataset = dataset.map(tokenize_function, batched=True)

train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_rows))
test_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_rows))
```

```
Map: 100%  40/40 [00:00<00:00, 760.80 examples/s]
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)

peft_model.print_trainable_parameters()
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```

```
from transformers import TrainingArguments, Trainer

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)
```

```
trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[5/5 01:25, Epoch 1/1]

Step	Training Loss

```
TrainOutput(global_step=5, training_loss=0.693804121017456, metrics={'train_runtime': 125.4387, 'train_samples_per_second':
0.319, 'train_steps_per_second': 0.04, 'total_flos': 5389576273920.0, 'train_loss': 0.693804121017456, 'epoch': 1.0})
```

```
trainer.evaluate(eval_dataset=test_dataset)

text = "The movie was fantastic and inspiring!"
inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)
prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[5/5 00:25]

```
Predicted label: 1
```

Start coding or generate with AI.